
The Cost of Autonomous Coordination: A Case Study of Multi-Agent Task Decomposition on Web Research

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Abstract

Multi-agent systems suggest improved performance on complex tasks through parallel decomposition and specialized coordination. Autonomously decomposing systems where a lead agent dynamically decides whether and how to spawn subagents should theoretically achieve at least single-agent performance by choosing not to decompose when coordination would hurt. We test this through a controlled evaluation comparing autonomous multi-agent decomposition against a matched single-agent baseline on web research benchmarks, using Claude Haiku 4.5. Across 400 runs on WebWalkerQA and BrowseComp-Plus, we find no statistically significant accuracy differences (McNemar’s $p = 0.82$, $p = 0.36$) while multi-agent systems incur 1.9–3.2 $\times$ higher costs. The single-agent system is either optimal or equal in 77% of paired comparisons. These findings suggest that effective multi-agent coordination may require architectural scaffolding rather than autonomous decomposition alone, though the results are specific to Claude Haiku 4.5 and should not be assumed to generalize to other model families or scales.

1. Introduction

Large language models now operate as agents capable of multi-step reasoning, tool use, and adaptive strategy refinement (Wang et al., 2024). As task complexity has grown, the field has increasingly turned to multi-agent systems (MAS), based on the intuition that specialized collaboration should outperform single-agent systems (Guo et al., 2024). This intuition appears well-founded: Anthropic’s multi-agent research system achieved 90.2% performance gains over single-agent baselines by decomposing queries into parallel subagent tasks (Anthropic Engineering, 2025), and Ope-

nAI’s Deep Research agent solves $\sim 51\%$ of BrowseComp problems that conventional models struggle with (Wei et al., 2025).

Yet mounting evidence contradicts this optimism. Multi-agent advantages diminish substantially with more capable base models (from $\sim 10\%$ gains to just 3%) while incurring 4–220 $\times$ higher token costs (Gao et al., 2025). Systematic analysis has identified 14 distinct multi-agent failure modes (Cemri et al., 2025), and recent controlled experiments show that coordination overhead can cause multi-agent variants to *degrade* performance by 39–70% on sequential reasoning tasks (Kim et al., 2025).

This tension is especially strong for *autonomously decomposing* systems, architectures where a lead agent dynamically decides whether and how to spawn subagents rather than follow a fixed pipeline. Such systems power popular production tools like Claude Code, suggesting the ability to capture multi-agent benefits when appropriate while avoiding overhead when not. In principle, autonomous decomposition is often assumed to weakly dominate single-agent execution, since the system can always choose minimal decomposition. However, this guarantee assumes that the presence of decomposition capability does not itself alter agent behavior when unused. We empirically test this assumption on Claude Haiku 4.5.

Problem and Hypothesis. Despite widespread deployment, no controlled comparison has isolated the effects of autonomous decomposition from confounding implementation differences. We address this gap with a falsifiable hypothesis: *if autonomous decomposition provides genuine value, we should observe higher accuracy at similar cost, or similar accuracy at lower cost, compared to a matched single-agent baseline.*

We evaluate one specific instantiation using the Claude Agent SDK with Claude Haiku 4.5 on web research benchmarks. Both conditions use identical base architectures, tools, and budgets; they differ in whether subagent spawning is enabled and in minor adaptations to Anthropic’s published system prompts (Anthropic, 2025) necessary for the single-agent baseline.

Contributions. We provide (1) a controlled comparison

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of one centralized autonomous decomposition architecture against a matched single-agent baseline on web research tasks, (2) empirical characterization of behavioral patterns associated with success and failure in this system, and (3) practical observations relevant to deployment decisions under realistic cost constraints.

Overview. We evaluate on a selection of question-answer pairs from WebWalkerQA (Wu et al., 2025) (iterative web traversal) and BrowseComp-Plus (Wei et al., 2025) (hard-to-find information retrieval). We measure accuracy, cost, and behavioral dynamics to understand not just *whether* decomposition helps in this configuration but what patterns correlate with success or failure.

Summary of Findings. Our results do not support the hypothesis that autonomous decomposition improves outcomes. Across 400 runs, accuracy differences between systems are not statistically significant, while multi-agent costs are $1.9\text{--}3.2\times$ higher. The single-agent system Pareto-dominates in 77% of paired comparisons. The autonomous system exhibits two distinct patterns: on structured tasks, it collapses to minimal decomposition while still incurring overhead; on open-ended tasks, it frequently explodes into aggressive parallelization that correlates with failure. These findings suggest that, for this model and architecture, effective multi-agent coordination may require explicit scaffolding rather than autonomous decision-making.

2. Related Work

Web Research Benchmarks. BrowseComp (Wei et al., 2025) comprises 1,266 problems requiring persistent, creative search across the open web. Questions are constructed so that shallow retrieval fails; GPT-4o achieves only 1.9% with browsing, while Deep Research reaches $\sim 51\%$. BrowseComp-Plus (Chen et al., 2025) extends this with retrieval-specific evaluation and query auditing. WebWalkerQA (Wu et al., 2025) instead evaluates systematic traversal within structured site hierarchies (680 questions across 1,373 pages), emphasizing methodical navigation over creative search reformulation. We select BrowseComp-Plus for its difficulty and independently verified question-answer pairs, and WebWalkerQA for its relative efficiency and structural simplicity. Together, these benchmarks span a spectrum from open-web deep research to structured site traversal.

Multi-Agent Architectures. MAS coordination structures include independent (ensemble aggregation), decentralized (peer-to-peer debate) (Du et al., 2023), centralized (orchestrator-worker) (Hong et al., 2024), and hybrid topologies (Guo et al., 2024). The centralized orchestrator-worker pattern is common in production systems: Anthropic’s agen-

tic research system uses a Claude Opus 4 lead coordinating Sonnet 4 subagents (Anthropic Engineering, 2025). A critical distinction exists between *fixed* pipelines with predetermined decomposition and *autonomous* systems where the lead dynamically decides whether to decompose. The latter powers Claude Code and is our experimental focus.

Single vs. Multi-Agent Comparisons. Early work reported substantial multi-agent advantages (Li et al., 2024; Qian et al., 2025), but recent evaluations are more nuanced. Gao et al. (2025) find advantages diminish with capable models while costs increase dramatically. Cemri et al. (2025) identify systematic failure modes spanning node-level bottlenecks, edge-level information overload, and path-level error propagation. Most relevant to our work, Kim et al. (2025) demonstrate that coordination benefits are task-contingent: centralized coordination improves parallelizable tasks by 80.9% but *degrades* sequential reasoning by 39–70%. Their framework predicts optimal architecture for 87% of held-out configurations based on task properties.

Positioning. We evaluate one specific instantiation of autonomous decomposition (an orchestrator-worker pattern with Claude Haiku 4.5) isolating the mechanism by which a lead agent decides whether to spawn subagents. Unlike comparisons of fixed SAS vs. MAS pipelines, we test whether this decomposition capability adds value under realistic constraints.

3. Methods

3.1. System Architecture

Both systems use Claude Haiku 4.5 via the Claude Agent SDK package. We selected this platform for production robustness and reproducibility; Anthropic’s published multi-agent research system (Anthropic Engineering, 2025) provides documented implementation and prompt specifications (Anthropic, 2025) that we adapt for our evaluation. The two architectures differ in whether subagent spawning is enabled and in 8 minor prompt modifications removing references to multi-agent coordination. Changes involve removing phrases such as ‘working as part of a team,’ ‘provided by a lead agent,’ and ‘to the lead researcher’ (full prompt diff in supplementary materials).

Single-Agent Baseline. The single-agent system executes an agentic loop for up to 50 turns with four tools: WebSearch (broad queries), WebFetch (full page retrieval), Read (local files), and Bash (system utilities). The agent alternates between tool execution and reasoning until producing a final answer or exhausting its turn budget.

Agentic System. The autonomously decomposing multi-agent system adds a single Task tool enabling the lead agent to spawn subagents. Subagents receive the same four

tools as the single-agent baseline (without `Task` to preventing recursion) and operate under a 25-turn budget. The lead agent retains full autonomy over decomposition decisions: how many subagents to spawn, what tasks to delegate, and when to synthesize results.

Design Philosophy. We deliberately impose minimal constraints on the multi-agent system. Rather than forcing decomposition with prescribed subagent counts, we grant the lead agent complete autonomy, reflecting realistic deployment scenarios where systems must self-determine coordination. High tool budgets (50 turns for lead and 25 calls per subagent) are set to prevent artificial resource constraints.

3.2. Datasets and Evaluation Protocol

Benchmarks. We evaluate on an arbitrary selection of 100 questions from two complementary web research benchmarks. **WebWalkerQA** (Wu et al., 2025) (680 questions) emphasizes systematic traversal of structured website hierarchies, requiring multi-hop navigation through subpages. **BrowseComp-Plus** (Chen et al., 2025) (830 questions) tests extensive search for hard-to-find information across the open web. To preserve equivalence between the two architectures, we evaluate BrowseComp-Plus on the open web with a standard search implementation. Together, these span the spectrum from structured navigation to open-ended deep research.

Execution. Runs execute in parallel (up to 5 concurrent to respect rate limits), each in an isolated temporary directory to prevent cross-contamination. Both architectures receive identical questions under matched conditions.

Grading. We employ LLM-as-judge evaluation at two levels. At the *system level*, GPT-5-mini receives the question, agent response, and ground-truth answer, producing binary accuracy (yes/no) and a 0–10 correctness score capturing partial credit. At the *subagent level* (multi-agent only), Gemini-3-Flash assesses individual subtask completion (0–1) and quality (0–10), enabling analysis of where coordination succeeds or fails within the pipeline.

3.3. Metrics

We extract 25 metrics per run across five categories: **Performance** (binary accuracy, correctness score), **Resources** (cost in USD, execution time), **Activity** (turns, messages, total tool calls, per-tool breakdowns), **Coordination** (subagent count, lead/subagent message splits, task similarity via sentence embeddings), and **Errors** (tool call failures, API errors, timeouts).

Table 1. Performance comparison across benchmarks. Multi-agent achieves comparable accuracy at substantially higher cost. #Sub. indicates the mean number of subagents spawned per run.

Benchmark	System	Acc.	Cost	Time	#Sub.
WebWalker	Single	50%	\$0.24	89s	—
WebWalker	Multi	48%	\$0.47	128s	1.4
BrowseComp-Plus	Single	13%	\$0.85	108s	—
BrowseComp-Plus	Multi	18%	\$2.72	450s	4.9

3.4. Statistical Analysis

For binary accuracy, we use McNemar’s test over discordant pairs (questions where exactly one system succeeded) to assess whether differences exceed chance. For continuous metrics, we report paired mean differences with 95% bootstrap confidence intervals (10,000 resamples). Effect sizes (Cohen’s *d*) quantify practical significance. All analyses treat the question as the unit of pairing, enabling direct per-question comparison between architectures.

4. Results

We evaluate 400 runs: 100 each for single-agent and multi-agent conditions on both WebWalkerQA and BrowseComp-Plus. All results use paired comparisons where applicable.

4.1. Overall Performance

Table 1 presents performance across conditions. On WebWalkerQA, the single-agent system achieves a negligible difference of 50% accuracy versus 48% for multi-agent. On BrowseComp-Plus, multi-agent shows a modest advantage (18% vs. 13%), though both systems struggle with this challenging benchmark. These accuracy differences come at substantial cost: multi-agent systems consume $1.9\times$ (WebWalker) to $3.2\times$ (BrowseComp-Plus) more resources, with execution times increasing $1.4\times$ and $4.2\times$ respectively.

Cost per correct answer highlights the efficiency gap: single-agent achieves \$0.49/correct on WebWalker versus \$0.98 for multi-agent ($2.0\times$), and \$6.52/correct on BrowseComp-Plus versus \$15.11 ($2.3\times$). The multi-agent system consistently requires more resources without proportionate accuracy gains.

Figure 1 illustrates the relationship between execution cost and run-level accuracy. Multi-agent systems exhibit substantially higher variance in cost across both benchmarks, especially in failed runs. On BrowseComp-Plus, successful multi-agent runs tend to incur higher median cost than single-agent runs, while failures span a wide cost range. This tendency holds for WebWalkerQA, although the cost distributions of those runs include more overlap. Overall, increased execution cost does not translate into higher accu-

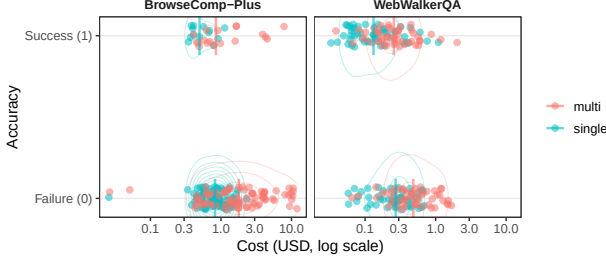


Figure 1. Run-level accuracy versus execution cost for single-agent and multi-agent systems on BrowseComp-Plus (left) and WebWalkerQA (right). Each point corresponds to one run. Vertical jitter is applied. Contours indicate kernel density estimates within each outcome region. Vertical ticks denote the median cost for each system, computed separately for successful and failed runs.

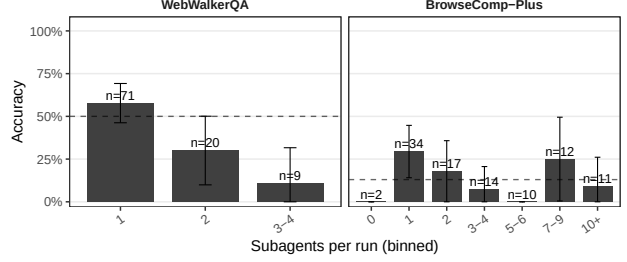


Figure 2. Accuracy as a function of the number of subagents per run (multi-agent only), shown separately for WebWalkerQA (left) and BrowseComp-Plus (right). Bars report mean accuracy within each subagent-count bin; error bars denote 95% confidence intervals. Dashed horizontal lines indicate the corresponding single-agent baseline accuracy for each benchmark. Sample sizes per bin are annotated above each bar.

racy.

4.2. Paired Statistical Comparison

Neither accuracy difference reaches statistical significance. McNemar’s test on discordant pairs (questions where exactly one system succeeded) yields $p = 0.82$ for WebWalker (11 single-agent wins vs. 9 multi-agent wins among 20 discordant pairs) and $p = 0.36$ for BrowseComp-Plus (7 vs. 12 among 19 discordant pairs). Bootstrap 95% confidence intervals for accuracy differences span zero in both cases: WebWalker $[-0.11, +0.07]$, BrowseComp-Plus $[-0.04, +0.13]$.

Effect sizes (Cohen’s d) confirm negligible accuracy differences ($d = 0.04$ WebWalker, $d = -0.14$ BrowseComp-Plus) but large cost differences ($d = -0.74$ and $d = -0.92$ respectively, where negative indicates multi-agent higher).

Single-Agent Efficiency. Across paired comparisons, the single-agent system is simultaneously cheaper and achieves equal or better accuracy in 77% of cases (154/200): 78% on WebWalker, 76% on BrowseComp-Plus. This pattern—where single-agent performs at least as well on accuracy while incurring lower cost—characterizes the typical outcome in our evaluation.

4.3. Decomposition Dynamics

The autonomous system exhibits strikingly different decomposition behavior across benchmarks.

WebWalker: Self-Pruning. On WebWalkerQA, the multi-agent system converges to minimal decomposition: 71% of runs spawn exactly one subagent, 20% spawn two, and only 9% spawn three (the observed maximum). Accuracy correlates negatively with subagent count: 57.7% for single-subagent runs, 30.0% for two subagents, and 11.1% for three ($r = -0.32$, $p = 0.001$). However, we cannot determine causality as the system may spawn more subagents precisely when it perceives tasks as more difficult.

BrowseComp-Plus: Coordination Explosion. On BrowseComp-Plus, decomposition ranges from 0–29 subagents (median: 2, mean: 4.9). The system frequently attempts aggressive parallelization, with 33% of runs spawning 5+ subagents and 11% spawning 10+. Accuracy patterns by subagent count are suggestive but rest on limited samples: single-subagent runs achieve 29.4% accuracy (10/34, the highest observed) while runs with 4–6 subagents achieve 0% (0/14) and high-intensity runs (10+ subagents) yield 9.1% (1/11). These subgroup analyses should be interpreted cautiously given sample sizes.

Figure 2 summarizes the relationship between subagent count and task accuracy across both benchmarks. On WebWalkerQA, performance degrades monotonically as additional subagents are introduced, with all multi-agent configurations underperforming the single-agent baseline. BrowseComp-Plus shows a non-monotonic pattern: while small multi-agent configurations (1–2 subagents) match or slightly exceed the baseline, intermediate configurations degrade substantially, and higher-count regimes (7–9 subagents) recover some performance despite increased variance. These results suggest that the effect of agent decomposition is highly task-dependent and that increasing the number of subagents does not reliably yield accuracy gains.

4.4. Behavioral Patterns

Two behavioral patterns emerge from the activity metrics, though neither is necessarily causal.

Depth vs. Breadth. We define *depth* as the ratio of WebFetch calls (retrieving full page content) to WebSearch calls (issuing queries). Successful runs exhibit significantly higher depth than failures across conditions. The effect is strongest on BrowseComp-Plus: correct runs average depth 0.29 versus 0.13 for incorrect runs ($t = 5.22$, $p < 0.0001$). This correlation suggests that

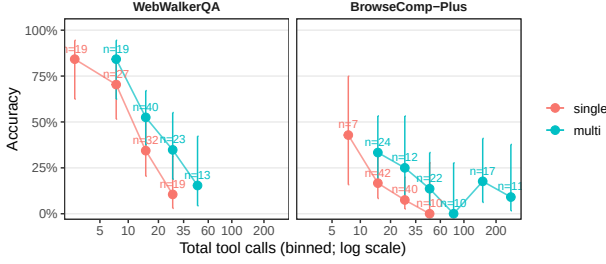


Figure 3. Activity correlates with failure (“thrashing”). Binned accuracy vs. total tool calls (pseudo-log) for single- and multi-agent systems on WebWalkerQA and BrowseComp-Plus. Markers are bin means (shared bins); error bars are 95% Wilson intervals; labels show n runs/bin. Higher activity is associated with lower accuracy in both benchmarks.

reading retrieved content carefully, rather than issuing many queries with shallow follow-through, is associated with success.

Activity-Accuracy Correlation. Activity metrics correlate negatively with accuracy across all conditions: total tool calls ($r = -0.24$ to -0.36), search calls ($r = -0.27$ to -0.34), cost ($r = -0.27$ to -0.33), and time ($r = -0.21$ to -0.36), all $p < 0.05$. Runs that consume more resources tend to fail (Figure 3). This pattern, which we term *thrashing*, admits two interpretations: (1) escalating effort fails to recover from early missteps (causal), or (2) difficult questions independently induce both high activity and low accuracy (confounded).

Error Propagation. Multi-agent systems encounter substantially more errors (counted as tool failures): 4.88 errors/run on BrowseComp-Plus (vs. 0.91 single-agent) and 1.46 on WebWalker (vs. 0.86). These errors accumulate across subagents, with 77% of BrowseComp-Plus multi-agent runs experiencing at least one error compared to 63% of single-agent runs.

5. Discussion

Our evaluation tests whether one autonomous multi-agent decomposition compares favorably to a matched single-agent baseline. The theoretical expectation—that autonomous decomposition should weakly dominate single-agent execution, since it can choose not to decompose—is not supported by our results. This expectation assumes *architectural neutrality*: that the presence of decomposition capability does not alter behavior when unused. Our findings suggest this assumption may not hold.

5.1. Two Failure Patterns

Collapse (WebWalker). On structured navigation tasks, the system self-prunes to minimal decomposition: 71% of runs

spawn exactly one subagent, with a maximum of three. This conservative behavior suggests the lead agent may recognize that WebWalker’s sequential traversal tasks do not benefit from parallelization. Yet even minimal coordination incurs $1.9\times$ cost overhead without accuracy benefit. This overhead may arise from fixed costs of the coordination infrastructure (e.g. prompt, spawning, and synthesis overhead) rather than from poor decomposition decisions.

Explosion (BrowseComp-Plus). On open-ended research tasks, the system frequently attempts aggressive parallelization, with runs spawning up to 29 subagents. Accuracy correlates negatively with subagent count, though this pattern is consistent with either (a) over-decomposition causing failure through information fragmentation and error compounding, or (b) the system spawning more subagents on harder questions that would fail regardless. The $5\times$ higher error rate in multi-agent BrowseComp-Plus runs is consistent with both interpretations.

One interpretation unifying both patterns is that the lead agent lacks effective meta-cognition about when decomposition helps. On WebWalker, it pays coordination overhead for minimal benefit. On BrowseComp-Plus, it appears to decompose aggressively without corresponding synthesis capacity. However, alternative explanations include fixed architectural overhead, prompt-induced behavioral changes, or task difficulty confounds.

5.2. The Depth-Breadth Correlation

The depth metric (fetch-to-search ratio) correlates strongly with success: on BrowseComp-Plus, correct runs achieve depth 0.29 versus 0.13 for failures ($p < 0.0001$). Multi-agent systems, which issue $3.3\times$ more searches on BrowseComp-Plus, exhibit lower depth on average. This pattern is consistent with, but does not prove, breadth-first strategies fragmenting attention without improving outcomes.

Anthropic’s multi-agent research system explicitly prompts for depth-vs-breadth decisions (Anthropic Engineering, 2025). Our autonomous system, granted freedom to determine its own strategy, may default to breadth-first escalation. Whether architectural constraints enforcing depth would improve outcomes remains untested.

5.3. Implications for Deployment

For practitioners deploying systems similar to ours (Claude Haiku 4.5, centralized agent system, web research tasks), our findings suggest several considerations. First, single-agent architectures provided better cost-efficiency in our evaluation, though this may not generalize to other domains. Second, explicit escalation strategies such as the agent cascade proposed by Gao et al. (2025) may outper-

form autonomous decomposition, though we did not test this directly. Third, the negative correlation between activity and accuracy suggests that escalating resource consumption signals likely failure, which could inform early termination decisions. Finally, if deploying multi-agent systems, hard caps on subagent counts may prevent the coordination explosion pattern we observe on BrowseComp-Plus.

5.4. Implications for System Design

Anthropic’s production system achieves strong results through extensive scaffolding: detailed task descriptions, scaling rules, depth-vs-breadth heuristics (Anthropic Engineering, 2025). Our minimal-constraint design, intended to test whether good prompts suffice, suggests they may not. Whether more capable models could succeed with autonomous decomposition remains open. Our results also suggest that merely adding decomposition capability may alter system behavior and introduce fixed overhead, even when decomposition is minimally used. System designers should consider whether the coordination pathway itself affects performance.

5.5. Interpreting Our Findings: Scope and Limitations

Our findings are bounded by several important constraints that affect interpretation and generalizability.

Model Capability. Most critically, we evaluate only Claude Haiku 4.5. Our findings may not generalize to more capable models; autonomous decomposition failures may simply reflect insufficient capability. Budget constraints prevented systematic comparison across model tiers, but this remains a critical limitation. The cost barriers are substantial: our 400-run evaluation with Haiku cost approximately \$428 total (\$0.24 vs \$0.84/run for single-agent on WebWalker vs BrowseComp-Plus, \$0.47 vs \$2.72 for multi-agent). Replicating this evaluation with Claude Sonnet 4.5 would cost roughly \$1,000–2,000 based on the $3\times$ token pricing difference, while evaluating Claude Opus 4.5 on the entire BrowseComp-Plus corpus would likely exceed \$15,000.¹ This cost structure is one reason systematic multi-agent evaluation at the frontier remains uncommon in literature.

Causal Interpretation. Our correlational findings require careful interpretation. Systems that spawn many subagents may simply perceive tasks as difficult; considering accuracy by subagent count without controlling for task difficulty conflates decomposition and difficulty effects. The negative correlation between subagent count and accuracy should not be interpreted causally. Relatedly, many subgroup analyses rest on small samples: 11 runs with 10+ subagents, only 18

correct multi-agent runs on BrowseComp-Plus, and 5 correct runs with 2+ subagents. This makes claims about high-intensity multi-agent behavior suggestive and not definitive.

Evaluation Infrastructure. Web research benchmarks evaluate against a changing environment (we mitigate this through paired evaluation but cannot eliminate it), and we rely on LLM-as-judge evaluation without systematic human validation which may introduce bias (Zheng et al., 2023). Questions were selected sequentially from each benchmark rather than randomly sampled, which may introduce ordering biases if benchmark creators organized questions by difficulty or topic.

Generality. Our results should not be interpreted as demonstrating that multi-agent systems are inferior to single-agent alternatives *in general*. We test one specific configuration: autonomous orchestrator-worker decomposition with Claude Haiku 4.5 on web research tasks. Anthropic’s production multi-agent research system achieves strong results through extensive scaffolding (Anthropic Engineering, 2025); our minimal-constraint design tests whether such scaffolding is necessary for this model, and the answer appears to be yes. Whether more capable models could succeed with autonomous decomposition, or whether alternative coordination policies would perform differently, remains unknown.

5.6. Open Questions and Future Directions

Several critical gaps remain in our understanding of autonomous multi-agent decomposition.

Model Scaling. If autonomous decomposition failures stem from insufficient meta-cognitive capability, frontier models might succeed where Haiku fails. However, the cost barriers described above explain why systematic multi-agent evaluation at the frontier remains sparse in the literature. Pass@ k evaluation, which would substantially improve statistical power for analyzing runs on difficult benchmarks like BrowseComp-Plus, multiplies these costs by k . Despite these barriers, model scaling represents the most critical unknown: whether our findings reflect fundamental limitations of autonomous decomposition or merely the capabilities of our base model.

Causal Identification. Our observational design cannot distinguish whether decomposition causes failure or merely correlates with task difficulty. Controlled experiments that manipulate decomposition orthogonally to task assignment would enable causal identification. For instance, forcing the system to spawn specific subagent counts regardless of task properties, or comparing identical tasks under different decomposition constraints, could isolate decomposition

¹And this assumes comparable resource consumption; preliminary testing on Opus 4.5 resulted in a \$50+ run.

effects from difficulty confounds.

Alternative Coordination Strategies. Gao et al. (2025) propose agent cascade and routing strategies, and (Kim et al., 2025) evaluate four different fixed agentic coordination structures that could be directly compared against autonomous decomposition. Such comparisons would help determine whether the issue is autonomous decision-making specifically or multi-agent coordination more broadly.

Architectural Scaffolding. Architecturally determined systems (hard caps on subagent counts, required synthesis checkpoints, redundancy detection) could be compared against our autonomous baseline to quantify the value of scaffolding. Our findings suggest that Anthropic’s production system succeeds precisely because it provides such scaffolding; controlled ablations would test this interpretation.

Task-Specific Evaluation. Cross-benchmark evaluation on tasks with naturally parallel structure would test whether our findings generalize beyond retrieval-focused benchmarks or reflect task-specific dynamics. Kim et al. (2025) demonstrate that coordination benefits depend critically on task properties; extending their framework to autonomous decomposition could predict when such systems should succeed or fail.

5.7. Implications

For Deployment: (1) Consider single-agent defaults for cost-efficiency; (2) explicit escalation (Gao et al., 2025) may outperform autonomous decomposition; (3) monitor activity patterns as escalating consumption signals likely failure; (4) bound decomposition with hard caps.

For Design: Our minimal-constraint design suggests scaffolding may be necessary, at least for this model. Adding decomposition capability may alter behavior and introduce fixed overhead even when minimally used.

6. Conclusion

We present a controlled evaluation of one autonomous multi-agent decomposition architecture on web research tasks. Across 400 runs on WebWalkerQA and BrowseComp-Plus, we find no statistically significant accuracy differences (McNemar’s $p = 0.82$ and $p = 0.36$) while multi-agent systems incur $1.9\text{--}3.2\times$ higher costs. In 77% of paired comparisons, the single-agent system is simultaneously cheaper and achieves equal or better accuracy.

The autonomous system exhibits two patterns: collapse to minimal decomposition with coordination overhead on structured tasks, and aggressive parallelization correlating

with reduced accuracy on open-ended tasks.

These findings do not indict multi-agent systems generally. Anthropic’s production research system achieves strong results through extensive scaffolding. Our contribution is demonstrating that, for Claude Haiku 4.5 on web research, such scaffolding appears necessary: autonomy alone does not optimize decomposition.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

Reproducibility Statement

All code, evaluation data, prompts, and raw results will be made publicly available upon publication. Anonymous supplementary materials including full prompt specifications and per-run metrics are provided as supplementary materials.

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